Cross-Family and Partial-Family Weather Prediction

Common subdomain coordinates and the California–Florida construction

Thy Nguyen

29 October 2026
Mathematical methods manuscript, revised edition
DOI: 10.5281/zenodo.23785875

Abstract

A weather predictor fitted in one subdomain can be applied to another only after observations, targets, horizon and conditional predictive structure have been aligned. This paper develops a mathematical study for source-trained weather transfer, partial-family eligibility and target-specific compatibility. California and Florida label a two-location worked construction motivated by the author's geographic-transfer workflow; all numerical calculations here are stipulated examples, not reported experiments. The treatment distinguishes transporting a rain mapping between locations from predicting a different weather observable using only rain data. Conditional-law transport, support tests, concept-shift counterexamples, calibration rules and conservative decision gates establish an explicit application procedure.

Research lineage and mathematical status. This study develops a WPM mathematical extension with reference to the author's published WPM architecture [1] and foundational n/s scaling-spectrum framework [2]. The n/s paper concerns conditional moment-spectrum parameterization; the WPM paper provides the broader architectural reference. The results proved here are mathematical consequences of stated assumptions, usually applications of standard linear algebra, probability or optimization. The WPM contribution is their organization into an explicit workflow. Modeling constructions and applicability conditions are identified separately; no universal forecasting superiority is asserted.

Predictor inputs. The forecasting layer described here is data driven: it learns from observed histories and matured outcomes. It is not fed the system's true governing equation or unobserved future states at issuance. Physical equations used below define observables, explain a limiting example or specify a separately labeled comparator. Optional supplied-map inverse problems and physics-informed variants must be declared separately; they are not evidence that the default forecast knows the generator.

1 Two locations, several subfamilies and three different tasks

Let $d \in \{A, B\}$ denote California and Florida observation supports, respectively. A site is a station or declared grid support, not an entire state's climate. Let f denote a declared subfamily: for example a weather-event target, a seasonal regime, or a cluster defined from observation histories. State which meaning is used. Rain, strong wind and temperature transition are useful target-indexed branches; they are neither mutually exclusive nor a partition of atmospheric states.

There are three distinct transfer questions:

1. **Cross-location, same target:** fit a California rain predictor and evaluate Florida rain using Florida issuance-time observations.
2. **Cross-regime, same target:** fit a rain predictor on one declared atmospheric subfamily and apply it to another, with a valid representation and support check.
3. **Cross-target:** use information learned for rain to predict wind or temperature. This needs a target bridge, joint model or additional labeled data; it does not follow from sharing coordinates.

The geographic construction transports a learned source mapping rather than future California weather values into Florida. It need not assert that California weather causally determines Florida weather. The archived workflow motivates the separation between portable rain/wind mappings and potentially incompatible temperature-transition mappings. This study supplies rules and constructed examples without reproducing experimental tables or asserting universal geographic portability.

2 Write both native contracts and the common contract

At location d, let X_d be observed atmospheric channels and $Y^T_{d,t,P}$ the future target. Its measurement support, sampling interval, unit convention, sensor height, accumulation window and release timing belong to the contract. Temperature swing at the endpoint and maximum swing within an interval remain different targets after any normalization.

Define a system normalizer $N_{S,d}$ and target normalizer $N_{T,d}$:

$$Z_d = N_{S,d}(X_{d,\le t}, C_{d,\le t}), \qquad U_d = N_{T,d}(Y^T_{d,t,P}; C_{d,t}). \tag{2.1}$$

All normalizer parameters used at issuance are known or estimated before issuance. The transformation of the future label is evaluated after observation, but its rule is frozen beforehand. S may include location, season, geometry, measurement support and dynamical time, not just a scalar scale.

The shared predictor is $g_{T,P}(Z_d)$ for the common U contract. If an output is transformed back to physical units, use the destination inverse map. A dimensionless horizon based on a characteristic clock changes the physical endpoint unless it is explicitly mapped back. For a fixed 24-hour query, both locations retain 24 elapsed hours regardless of their different variability scales.

Application rule. Specify the source and destination support, target, horizon, normalizers and permitted destination data before calling a mapping cross-family. Matching feature-vector lengths is insufficient.

3 Seasonal common coordinates for California and Florida

For channel j, fit a month/hour or smooth seasonal reference from a permitted historical period:

$$Z_{d,j}(t) = \frac{X_{d,j}(t) - \mu_{d,j}(c_t)}{s_{d,j}(c_t)}, \qquad s_{d,j}(c_t) > 0. \tag{3.1}$$

In practice use a declared scale floor and pooling for sparse strata. The reference climatology may use destination historical observations without fitting the destination prediction coefficients.

That is destination-specific representation fitting, and must be disclosed. It is not a claim of having no destination data. Known unit conversions alone require less information than an estimated local climatology.

Worked example 3.1 (Same normalized state, different native measurements)**.** Take stipulated reference temperature means $(18, 27)$ degrees C and standard deviations $(4, 2)$ for California and Florida. Observed values $(22, 29)$ map to $Z_T = 1$ in both locations. A shared model issuing a normalized endpoint mean .5 maps to physical means 20 and 28 degrees C. These arithmetic values describe coordinate conversion only; they establish no equality of future laws.

Additional common channels may include pressure anomalies, moisture deficit, recent precipitation, vector wind, cloud changes and short/long variability ratios. Each channel must have the same meaning at both locations. A source feature computed from a dense station network cannot be replaced by an unavailable destination measurement without a missing-feature strategy. Missingness can itself shift the conditional law.

For a relative event, define a local threshold from a reference distribution of the *same future-window functional*. For nonnegative accumulation $A_{d,t,P}$ and season c_t, let $b_{d,T,P}(c_t)$ be a pre-estimated quantile, such as the .85 quantile. Then

$$U_{d,t,P} = \mathbf{1}\{A_{d,t,P} > b_{d,T,P}(c_t)\}. \tag{3.2}$$

The strictness of the inequality and handling of ties must be fixed, particularly for dry-season zero mass. A continuous reference has probability .15 above its .85 quantile; a discrete precipitation reference need not. A percentile threshold defines relative unusualness, not a common physical amount.

Worked example 3.2 (Relative and absolute rain are different queries)**.** Suppose the predeclared .85 thresholds are 4 mm and 12 mm at the two locations. Accumulations 6 and 14 mm both satisfy the relative event. The same 6 mm amount is relatively extreme only at the first location. If the application instead asks for more than 10 mm anywhere, retain the 10 mm physical target and let its local base rate differ.

4 The source-trained resolution and pocket state

For each (W, m), build normalized observation histories with physical cadence and causal context. Fit a source expert $q_{A,T,P,W,m}$. A source-trained ensemble can be deployed as

$$\widehat{p}_{A\to B,T,P}(t) = \sum_k \alpha_{A,T,P,k}(Z_B(t)) q_{A,T,P,k}(Z_B(t)). \tag{4.1}$$

This formula is defined only for eligible inputs and a common binary target. Source coefficients remain frozen in a strict parameter-transfer branch. If destination labels alter q, α or a calibration map, that is an adapted branch.

Pocket descriptors retain existence, full flow cross-sections, location, signed residual behavior and the relational graph of independent pockets. A source pocket's ID is not automatically a matching destination pocket. Match interpretable descriptors or align a permutation-invariant set. A missing destination pocket can be a genuine absence, a sampling limitation or an unsupported forecast; distinguish these states rather than filling them with zero utility.

At location d, historical losses at issuance s enter a landscape at current time t only after the complete label and reporting delay have matured. A source-trained predictor may have a well-populated source landscape but no destination loss history. It must then report destination

reliability as unmeasured rather than reuse source calibration as an established destination property.

Flow geometry and position can support compatibility, but similarity of pockets alone does not prove conditional predictive equivalence. Likewise a future pocket projection is a model output from current information, not access to a destination's actual future landscape.

5 A sufficient theorem for geographic prediction transfer

Let P_A and P_B denote the distributions of normalized features Z and common target U, with all context and horizon included in Z. Let $P_B^Z \ll P_A^Z$ on the intended deployment support.

Theorem 5.1 (Conditional-law transport)**.** *If $P_A(U \mid Z) = P_B(U \mid Z)$ almost everywhere under P_B^Z, and $\ell(U, g(Z))$ is integrable, then with $w = dP_B^Z/dP_A^Z$,*

$$R_B(g) = \mathbb{E}_B[\ell(U, g(Z))] = \mathbb{E}_A[w(Z)\ell(U, g(Z))]. \tag{5.1}$$

For binary U under Brier loss, $\eta_A(Z) = \mathbb{P}_A(U = 1 \mid Z)$ is also destination optimal under those assumptions.

Proof. Condition destination loss on Z. The common conditional law makes that conditional expectation identical to the source conditional expectation. Change the feature measure from P_B^Z to P_A^Z using w and apply the tower property. The Brier optimum follows by expanding the conditional risk as $\eta(1-\eta) + (g-\eta)^2$. □

Equal normalized means and variances do not imply the theorem's conditional-law assumption. Importance weights can correct feature-distribution shift under the assumption; they cannot correct an unknown new relationship between target and features. Extreme weights also create high sampling uncertainty. Trimming weights changes the estimand and must be reported.

Proposition 5.2 (A bounded Brier transfer discrepancy)**.** *For binary labels and $g(Z) \in [0,1]$, let $\eta_d(z) = \mathbb{P}_d(U = 1 \mid Z = z)$. On overlapping feature support,*

$$\left| R_B(g) - \mathbb{E}_{Z \sim P_B^Z} \mathbb{E}_A[(g(Z) - U)^2 \mid Z] \right| \le \mathbb{E}_B |\eta_B(Z) - \eta_A(Z)|. \tag{5.2}$$

Proof. The conditional Brier risk equals $g^2 + \eta_d(1 - 2g)$. Subtract the two risks, use $|1 - 2g| \le 1$, then integrate the absolute bound. □

The right-hand side is not generally estimable from unlabeled destination features. It expresses the exact missing obligation when scale normalization has been performed but concept shift remains possible.

6 A counterexample: identical normalized weather, incompatible targets

Let $Z \in \{0, 1\}$ have equal probabilities at both locations. In the source let $U = Z$; in the destination let $U = 1 - Z$. The feature distributions, means, variances and support match perfectly. A source-optimal $g(Z) = Z$ has source Brier zero and destination Brier one; a destination forecast .5 has Brier .25. A domain classifier seeing only Z cannot detect this change because the feature laws are identical.

This proves that even exact normalized-feature alignment cannot establish a shared temperature-transition relationship. A global domain classifier answers whether features distinguish locations, not whether a particular target mapping is portable. Conversely, two locations can be easily distinguished by features yet have the same conditional rain law once relevant context is included.

Application rule. Make compatibility target- and horizon-specific. A successful rain transfer does not certify temperature transition, and a failed temperature transfer does not invalidate every wind forecast.

7 Learning from one subfamily: what must be shared

Let $f = a$ be the source subfamily and $f = b$ a destination subfamily, with common target U. Parameter transfer is justified by the same conditional-law and support conditions, now conditioned on subfamily membership. One may use a common feature map $Z = N_{S,d,f}(X)$ while retaining local metadata needed to identify the query.

Worked example 7.1 (Constructed common latent weather law)**.** Suppose a stipulated shared latent severity z satisfies $\mathbb{P}(U = 1 \mid z) = \sigma(-1 + 2z)$ in each of two normalized subfamilies, where σ is the logistic function. At $z = 0$, the event probability is $1/(1 + e)$; at $z = 1$, it is $e/(1 + e)$. Training only on subfamily a can identify this common response over its supported z range given adequate data. Application to b requires its z values to lie within source support and the conditional law actually to remain shared. The equality is an assumption of this example, not a fact implied by normalization.

For *different* targets, a possible bridge uses a latent variable H:

$$\mathbb{P}(U_b = 1 \mid X) = \int \mathbb{P}(U_b = 1 \mid H = h, X)\, dP(H = h \mid X). \tag{7.1}$$

Learning the rain branch alone may inform $P(H \mid X)$ but does not identify the wind-specific conditional factor. It requires paired cross-target labels, an externally justified measurement/response model, or other identifiable constraints. Supplying a response equation would be a declared bridge, distinct from the observation-only default.

Proposition 7.2 (Rain-only nonidentifiability)**.** *The joint distribution of observed X and rain label U_R alone does not identify $\mathbb{P}(U_V = 1 \mid X)$ for an unobserved wind label.*

Proof. Fix any distribution of (X, U_R). Extend it in model one with $U_V = 0$ almost surely and in model two with $U_V = 1$ almost surely. Both reproduce every available source observation and rain label, but they imply opposite wind probabilities. No estimator using only those observations can distinguish the models. □

Physical constraints can narrow this ambiguity if independently established, but the conclusion explains why source-only predictions for wholly unobserved targets must not be presented as established weather transfer.

8 Partial-family eligibility and component-wise predictions

For each output component j define an eligibility mask

$$M_{A\to B,j}(t, P) = \mathbf{1}\{\text{contract, channels, support and compatibility checks pass for } j\}. \tag{8.1}$$

Eligibility depends on the requested target and horizon and may vary with season or observation quality. It is a decision rule, not automatically the probability that the model is correct. Estimate any probabilistic interpretation against an independently defined correctness event.

An admissible output can be $(p_R, p_V, \perp_T)$: rain and wind transfer, temperature abstention. $\perp$ is not zero event probability. For an adapted temperature model, report $(p_R^{\text{source}}, p_V^{\text{source}}, p_T^{\text{adapted}})$ with distinct provenance. Do not average the three components into a common weather probability.

Within output j, eligible predictors may be combined using

$$\widetilde{\alpha}_{ij} = \frac{M_{ij}\alpha_{ij}}{\sum_k M_{kj}\alpha_{kj}}, \qquad \widehat{p}_j = \sum_i \widetilde{\alpha}_{ij} q_{ij}, \tag{8.2}$$

provided the denominator is positive and all q_{ij} predict the same event. Otherwise abstain or use a declared reference. Renormalizing invalid weights without a nonempty eligibility set is undefined.

Worked example 8.1 (Two components, two different decisions)**.** For rain candidates .3 and .6 with prior weights .4 and .6, if both are eligible, fusion gives .48. If only the first is eligible, it gives .3. If neither is eligible, no mixture exists. A temperature component can fail its own check at the same issuance without changing the rain target's definition.

9 Compatibility, calibration and stable adaptation

Fit source predictors first. On an earlier destination-development block, compare source transfer with a local reference under the same target and physical horizon. Use target-specific proper losses, class availability, support diagnostics, seasonal breakdown and data latency. A score normalized from these checks is a policy variable; it is not a universal constant or a proof of shared dynamics. When no destination labels are available, empirical destination calibration and target-conditional compatibility are unverified.

Separate four modes: source parameters with fixed unit conversion; source parameters with destination-estimated normalization; source parameters with a destination-fitted probability calibrator; and a destination-adapted target model. They use progressively different information and must retain separate labels. A destination-local benchmark is a comparison branch, not part of a strict source-only fit.

For frozen source score s, one calibration candidate is $\Gamma(s) = \sigma(a\,\mathrm{logit}(s) + b)$ with clipped $s \in (0, 1)$. Positive a preserves ranking, but finite-sample fitting can worsen probability accuracy. If ordering has reversed or a target relationship depends on missing context, an intercept/slope correction is inadequate. A calibrator cannot create evidence for an absent class from no labels.

Under a separately justified pure prior shift, where $P_B(Z \mid U) = P_A(Z \mid U)$ and only prevalence changes, Bayes' rule yields

$$\frac{\eta_B(z)}{1-\eta_B(z)} = \frac{\eta_A(z)}{1-\eta_A(z)} \frac{\pi_B/(1-\pi_B)}{\pi_A/(1-\pi_A)}. \tag{9.1}$$

This is an assumption-dependent correction, not a remedy for arbitrary geographic concept shift. For $\eta_A = .4$, $\pi_A = .2$ and $\pi_B = .4$, the destination odds become $16/9$ and the probability .64.

Enable a correction only under a frozen development rule, for example positive loss improvement over a practical margin in each chronological validation block. Compare on identical supported examples; a smaller evaluated set can otherwise mimic a gain. Retain the source or local reference when a correction fails, or abstain if neither is supported. Passing a gate in the past supplies no deterministic guarantee under future drift.

10 A complete two-location instructional construction

Step 1: name the observations and query. Let A be a California station and B a Florida station. Retain station support and physical hourly timestamps. Choose $P \in \{24, 48, 72\}$ hours. Specify accumulation-based rain, maximum-speed wind and interval temperature swing as separate target branches; these are instructional definitions, not a reconstruction of unspecified archive code.

Step 2: fit location-specific representation maps. Use only earlier local histories to estimate seasonal means, scales and any relative-event thresholds. Record their availability and fitting span. Convert units and wind direction consistently. Source-only fitting of response coefficients is compatible with local normalization but is not destination-data-free prediction.

Step 3: learn the source maps. Train the California lens bank and target-specific combination layer using matured source outcomes. Preserve each pocket's flow, position and existence; do not collapse independent regimes to one scalar. Store out-of-sample source histories used to train any higher layer.

Step 4: apply to destination histories. At Florida issuance t, compute $Z_B(t)$ and issue each same-target source prediction. The model uses Florida current and past measurements; it does not use future Florida labels or contemporaneous source future outcomes. Freeze any source mapping before the destination evaluation interval.

Step 5: check each branch independently. In a stipulated development worksheet, suppose rain and wind pass a common-support and target-specific compatibility policy while temperature swing fails. Issue rain and wind from the source-trained mapping and route temperature to a qualified local model or abstention. These choices illustrate partial transfer and are not numerical findings.

Step 6: keep absolute and relative queries separate. A local .85 threshold may align event unusualness, whereas a fixed millimeter, m/s or degree-C threshold serves a physical decision. If the relative target passes and the absolute target fails, retain that distinction rather than reporting a general weather-transfer success.

Step 7: update causally. Once full destination labels mature, update reliability records and, only if authorized by the protocol, normalization, calibration or adaptation. Record the change in information regime. A trained model may remain frozen while an eligibility decision changes; both need versioned audit records.

The opposite Florida-to-California direction is a separate task. Support, seasonal coverage, class balance and conditional laws may be asymmetric. A successful $A \to B$ application does not establish $B \to A$, let alone transfer to all climates.

11 What one-source learning can and cannot claim

If the source subfamily covers the normalized states and a shared conditional law holds, one fitted response map can predict the same target in another subfamily without new response coefficients. This is the mathematical mechanism behind a common representation. The prediction still needs the destination's issuance-time observations and its declared normalizer.

If destination states are outside source support, the response there is extrapolation. If only feature scales differ, normalization can help. If only probability scale differs, appropriate calibration may help. If the target-conditional response differs, representation repair, new target data or abstention may be necessary. These branches must remain distinct; repeated coordinate

enrichment does not prove that a failed response is learnable from the available source alone.

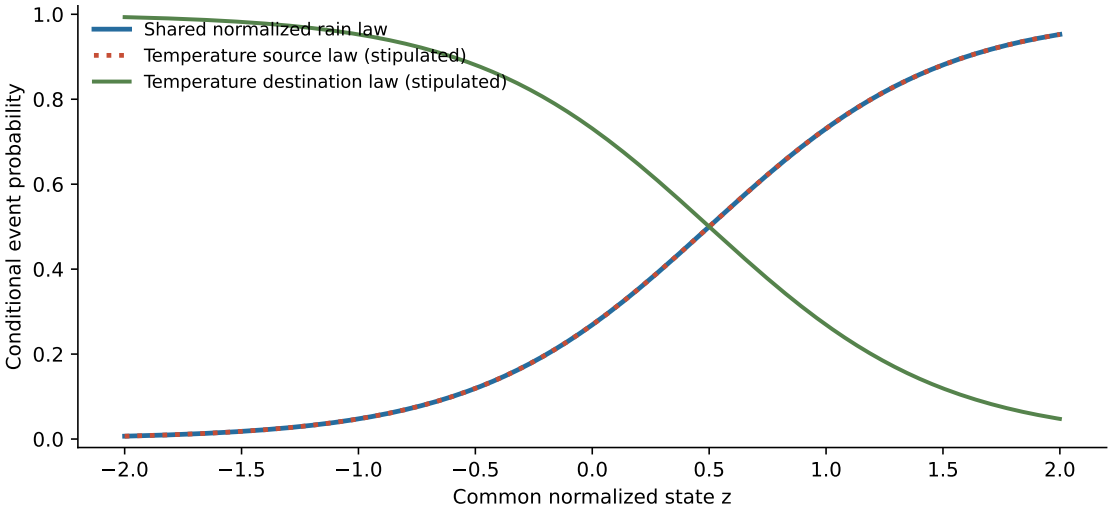


Figure 1: Constructed probabilities on a common normalized coordinate. A shared rain response illustrates compatible transfer; an inverted temperature response illustrates concept shift despite coordinate alignment. The curves are mathematical examples rather than fitted California or Florida records.

12 Proof trail and solved application questions

The transport theorem uses conditional expectation and a change of measure. The discrepancy bound is a direct Brier expansion. The concept-shift example proves that matching marginals is insufficient. The rain-only nonidentifiability proof uses two observationally identical extensions with opposite target laws. Bayes' rule yields the prior-shift correction under its stronger generative assumption. Together these results identify the exact assumptions needed before transferring a source weather map.

Exercise 12.1. At both sites, standardized temperature has mean zero and variance one. Is a shared future temperature-transition model established?

Solution. No. Those are feature moments, not a common conditional outcome law. The binary inversion counterexample fails even under identical full feature distributions.

Exercise 12.2. A model trained on California data uses Florida historical observations to estimate Florida climatology, but no Florida target labels to fit its response. Is it destination-data-free?

Solution. No. Its response parameters are source-trained, while its representation uses destination data. State both facts; neither invalidates the method.

Exercise 12.3. Rain and wind are eligible, temperature is not. May the output assign temperature risk zero and average all three probabilities?

Solution. No. Missing eligibility is abstention, not absence of an event. The targets are distinct and require component-wise outputs.

Scope of the mathematical exposition

Unless explicitly identified as a sourced evidence statement, numerical values are constructed arithmetic examples. Application requires specifying the observation process, fitting unknown quantities, and checking reliability for the intended population and horizon. The archival DOI for this study is 10.5281/zenodo.23785875.

References

[1] T. Nguyen. *The WPM Framework: A Generalized Mathematical Architecture for Cross-Scale, Spatial, and Dynamical Prediction*. Zenodo, 2026. doi:10.5281/zenodo.23248983.

[2] T. Nguyen. *An n/s Scaling-Spectrum Framework for Generalized Moment Scaling: Curvature, Signed-Moment Asymmetry, and Prospective Moment-Order Validation*. Zenodo, 2026. doi:10.5281/zenodo.23111783.

[3] T. Gneiting and A. E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association* 102(477), 359–378, 2007. doi:10.1198/016214506000001437.

[4] S. Boyd and L. Vandenberghe. *Introduction to Applied Linear Algebra: Vectors, Matrices, and Least Squares*. Cambridge University Press, 2018. https://stanford.edu/~boyd/vmls/.